

Notes on Adjusted R Squared

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ADJUSTED R SQUARED

Adjusted R Squared is a modified version of the R Squared metric that takes into account the number of predictors in a regression model.

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

R^2 - Goodness of fit
(greater is better)

Problem:
 $\hat{y} = b_0 + b_1X_1 + b_2X_2$ $\leftarrow$ $+ b_3X_3$

SS_{tot} doesn't change

Fig. 1. Illustration of Adjusted R^2 .

It is calculated using the formula:

$$\text{Adjusted } R^2 = 1 - \left(\frac{(1 - R^2)(n - 1)}{n - k - 1} \right)$$

where:

- R^2 is the coefficient of determination (R Squared),
- n is the number of observations,
- k is the number of predictors in the model.

The Adjusted R Squared value can be negative if the model fits the data very poorly. Unlike regular R^2 , which always increases when additional predictors are added, Adjusted R^2 penalizes the inclusion of unnecessary variables.

A higher Adjusted R^2 generally indicates a better-fitting model, while a lower value suggests that the predictors do not explain the variation in the response variable effectively. This makes Adjusted R^2 useful for comparing regression models that contain different numbers of predictors.

However, Adjusted R^2 should not be the only criterion used for model selection. Statistical significance tests, residual analysis, and domain-specific considerations should also be taken into account when evaluating model performance.